Optimizing Renewable Energy to Meet Malawi's Electricity Demand

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1. Introduction

Malawi's electricity sector continues to face persistent challenges related to supply insecurity, limited system flexibility, and heavy reliance on hydropower generation. Hydropower constitutes over 78% of the majority of installed generation capacity. This exposes the power system to significant climate-related risks, particularly recurrent droughts and hydrological variability. These vulnerabilities have resulted in frequent power outages, constrained reserve margins, and increased dependence on costly diesel-based generation to meet short-term supply deficits.

At the same time, electricity demand in Malawi is steadily increasing, driven by population growth, urbanisation, industrial development, and the Government's Malawi Rural Electrification Programme (MAREP). The National Energy Policy (2023) along-side Malawi Vision 2063 targets a significant expansion of electricity generation capacity to support these goals. However, the existing structure of the power system remains constrained and insufficiently equipped to accommodate rising demand while maintaining reliability and affordability.

In response, Malawi has prioritised diversification of the electricity generation mix, with a growing emphasis on renewable energy technologies, particularly solar photovoltaic (PV) generation. While solar PV offers substantial potential to complement hydropower and enhance energy security, its variability introduces operational challenges related to intermittency, peak demand management, and system balancing. Addressing these challenges requires investments in system flexibility solutions capable of supporting higher shares of variable renewable energy.

Battery Energy Storage Systems (BESS) have emerged as a critical enabling technology for power systems undergoing a clean energy transition. In 2024, Malawi commissioned its first-ever utility-scale BESS with a capacity of 20 MW, intended to support peak shaving, load shifting, frequency regulation, reserve provision, and enhanced integration of renewable energy. For Malawi, utility-scale BESS offers a significant opportunity to reduce reliance on diesel generation, phase out coal, strengthen grid resilience, and improve overall system reliability.

Against this background, this study applies the Open-Source Energy Modelling System (OSeMOSYS) to identify the least-cost and reliable renewable energy mix for meeting Malawi's electricity demand. The analysis focuses on assessing the role of renewable energy technologies and BESS under alternative development pathways, evaluating system costs, generation outcomes, and reliability implications.

1. Methodology

The analysis is conducted using the Modelling User Interface for OSeMOSYS (MUIO), a linear optimisation framework designed for long-term energy system planning and policy analysis. OSeMOSYS identifies the least-cost combination of generation, storage, and transmission technologies required to meet projected electricity demand while satisfying system constraints and policy objectives.

The modelling horizon covers the medium- to long-term evolution of Malawi's electricity sector, with demand growth assumptions informed by national development plans, historical electricity consumption trends, and projected economic and population growth. The years under study are 2022 to 2050. It is assumed that electricity demand grows by 6% to 8% over the modelling period, reflecting increasing access, industrial expansion, and rising per-capita consumption.

The model is calibrated using national electricity sector data, including installed generation capacity, technology costs, system efficiencies, and operational characteristics. Baseline calibration is undertaken from 2022 to 2024, representing historical data to ensure that the model accurately reflects the existing structure and performance of Malawi's power system.

Technologies represented in the model include hydropower, solar photovoltaic (utility-scale and distributed), diesel and Battery Energy Storage Systems. BESS is modelled to provide energy shifting, peak demand support, and system flexibility services that enable higher penetration of variable renewable energy sources.

Three scenarios are developed to explore alternative development pathways for Malawi's electricity sector:

Scenario 1: Business-as-Usual (BAU)

This scenario assumes the continuation of existing development trends, with limited diversification beyond currently planned generation projects. Hydropower remains the dominant source of electricity, while diesel generation is utilised to address supply shortfalls during periods of low hydrological availability. A 20 MW Battery Energy Storage System (BESS) is already installed, providing limited system flexibility.

Scenario 2: Renewable Energy Expansion without Storage

This scenario reflects increased investment in renewable energy, particularly hydropower and solar PV, in line with Malawi's Integrated Energy Plan. No additional battery energy storage capacity is introduced beyond the existing installed BESS. The scenario evaluates the implications of higher renewable energy penetration on system costs, reliability, and renewable curtailment in the absence of further storage-based flexibility.

Scenario 3: Renewable Energy Expansion with BESS Deployment

This scenario incorporates utility-scale Battery Energy Storage Systems alongside expanded renewable generation. BESS is deployed to support peak demand management, improve grid stability, and enhance the integration of solar PV. The scenario assesses the trade-offs between higher upfront capital investment and long-term system benefits, including reduced fuel costs, lower reliance on diesel generation, and improved reliability.

The results across scenarios are compared in terms of generation mix, total system costs, investment requirements, emissions implications, and system reliability indicators. This comparative analysis provides insights into the strategic role of BESS in strengthening Malawi's electricity system and supporting a resilient, low-carbon energy transition.

Malawi's OSeMOSYS Reference Energy System (RES)

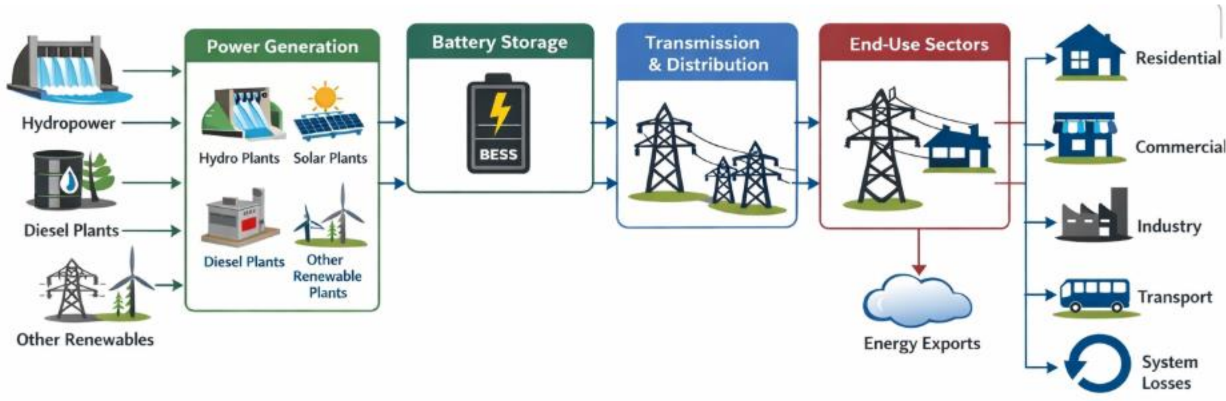

Figure 1 provides an overview of Malawi's current RES model based on OSeMOSYS

2. Results

The Reference Energy System (RES) illustrates the structure and flow of Malawi's OSeMOSYS energy system, from primary resource supply to final energy consumption. It incorporates key energy sources, including hydropower, solar PV, coal and diesel power, as well as electricity imported from the Southern African Power Pool (SAPP). These energy sources are transmitted through the national grid and distributed to final consumers across residential, commercial, and industrial sectors, reflecting national demand.

This study focuses on the evolution of electricity production by each power plant, the capacity allocation for each technology, relative investment costs, and associated CO_2 emissions. The results are summarized as follows:

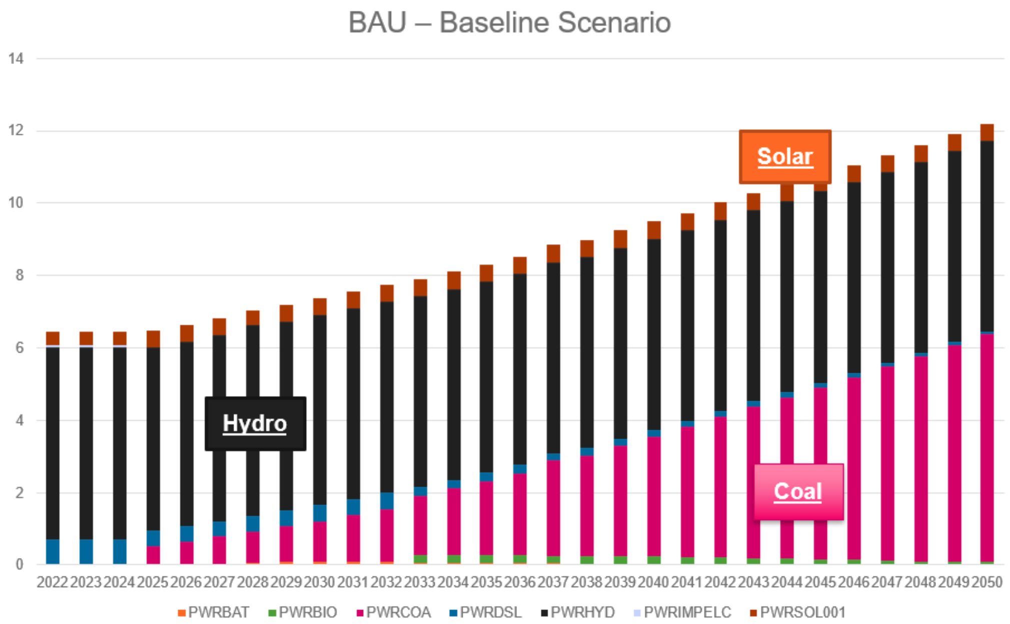

Figure 2: Production by Technology Mode (BAU)

In the Business-As-Usual (BAU) scenario, electricity production remains heavily reliant on hydropower plants, accounting for 78–85% of generation until 2040. Beyond this period, solar PV contributes only marginally, supplemented by diesel and a small share of coal-fired power plants.

However, maintaining hydropower at 78–85% of generation is plausible in the BAU context because Malawi's existing power system is already heavily hydro-dependent. If the optimisation assumes limited policy intervention, constrained investment, and slow technology uptake, the model will naturally continue to favour existing hydropower assets due to their low operating costs.

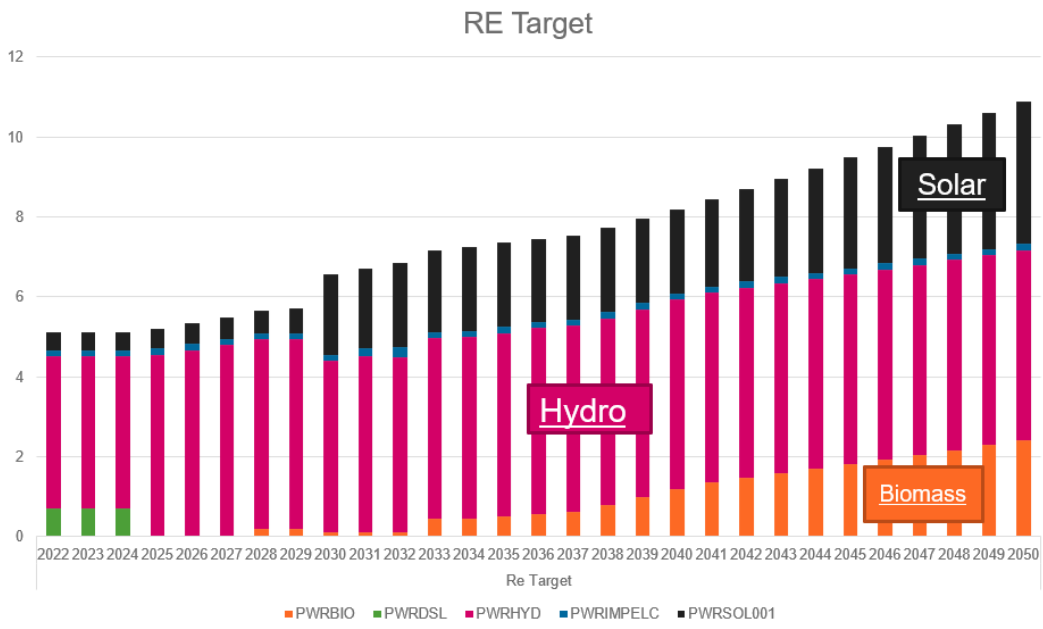

Figure 3: Production by Technology Mode (RE Target)

In contrast, the second scenario, which aligns with the renewable energy targets outlined in Malawi's Integrated Energy Plan, shows a significant increase in solar power generation and a sharp increase in capacity by 2023. This is accompanied by a rapid decline in diesel usage and the complete decommissioning of coal-fired power. This shift reflects the gradual replacement of fossil fuel-based generation. The model also indicates the potential role of biomass as an alternative renewable source, highlighting that while hydropower and solar PV are prioritized as the main renewable energy sources, biomass can contribute in the absence of fossil fuels.

Although OSeMOSYS does not explicitly account for barriers such as administrative hurdles or land availability which are particularly relevant for large-scale hydropower and solar PV deployment, these constraints can be indirectly represented in the model through delayed or constrained technology deployment. For example, while Malawi relies primarily on hydropower and solar, the model initially selects biomass as a cost-effective alternative technology, illustrating the flexibility of the system in integrating multiple renewable sources under different constraints.

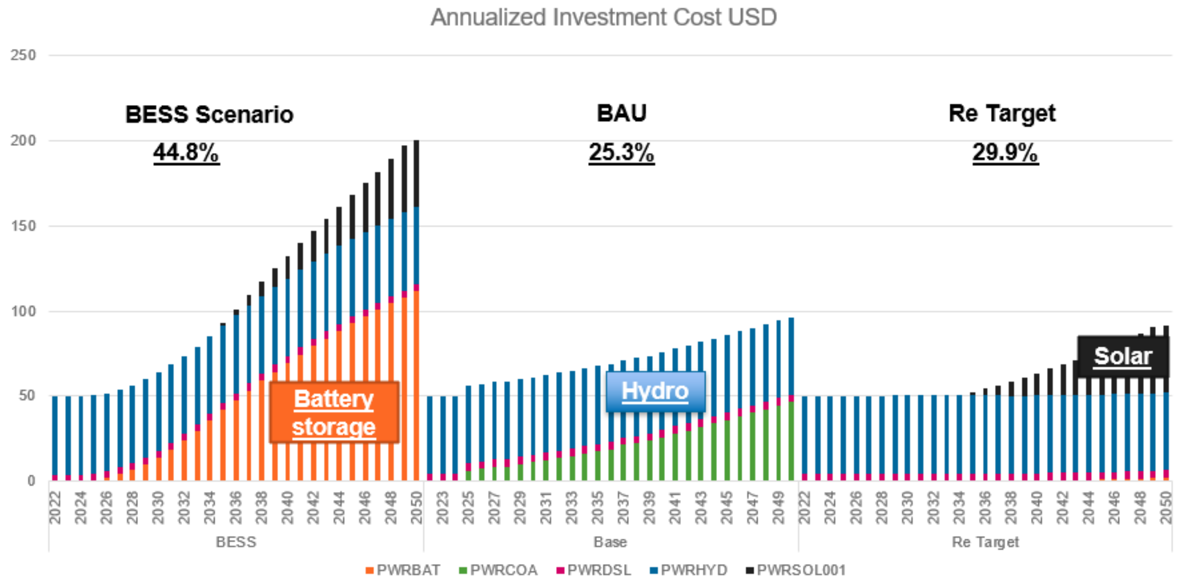

Figure 3: Annualized Investment Cost

This section presents the cumulative annualized investment costs for selected power generation technologies across three scenarios BESS, Base (Business-as-Usual), and Renewable Target from 2022 to 2050. The investment costs represent the capital expenditures required for the deployment and sustained operation of each technology over the modelling horizon.

In the BESS scenario, investment is heavily concentrated in battery storage and solar PV. BESS investment begins in 2025 and increases substantially, reaching approximately 111.55 by 2050, while solar PV investment starts in 2035 and grows to about 39.26 by 2050. No new investments are made in coal, and only minimal investments are observed for diesel, indicating reliance on storage-supported renewables. This outcome highlights BESS as the dominant cost driver in a high-renewable, low-carbon transition pathway.

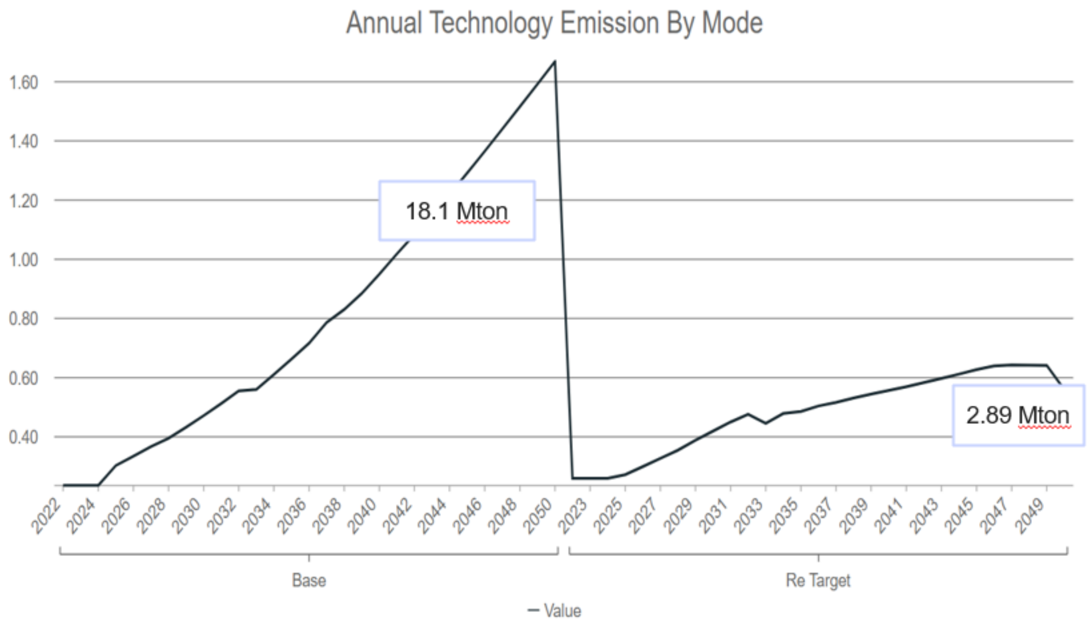

Figure 4: Emissions

When comparing all scenarios in terms of carbon emissions, the BAU scenario shows a significant use of coal and diesel. However, with the slowly phasing out of diesel and the decommissioning of coal by 2045, emissions are reduced by approximately 95%. Although the model retains a small diesel capacity, this has a negligible impact on Malawi's net-zero targets, as the country's main renewable energy sources primarily hydropower and solar PV effectively offset the remaining emissions.

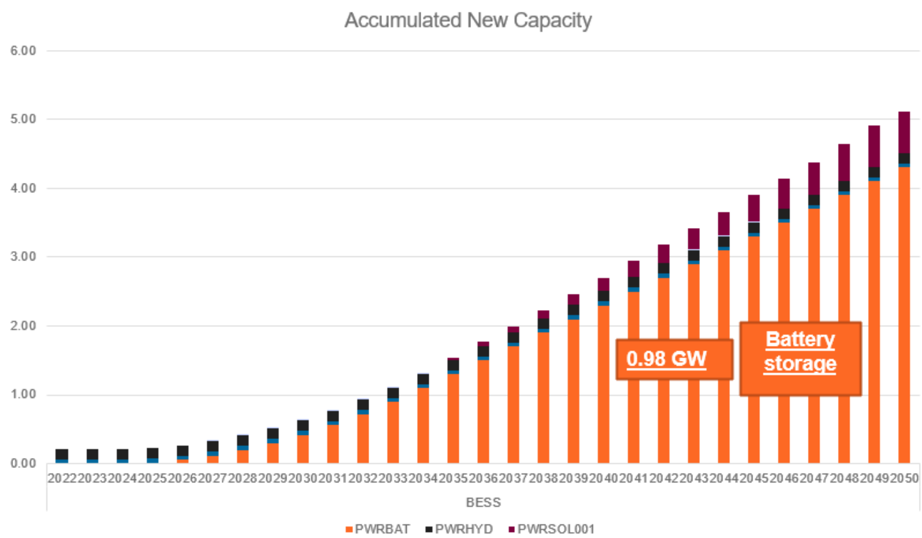

Figure 5: Accumulated New Capacity (BESS Scenario)

The third scenario introduces Battery Energy Storage System (BESS)/battery power (PWRBAT) technology into Malawi's power system. As a result, the deployment of BESS gradually increases over time, providing a mechanism to store excess electricity generated during periods of low demand and release it during peak demand. While the addition of BESS does not directly increase the overall generation capacity, it enhances the reliability and stability of the power system by effectively managing supply fluctuations. In consequence, the cumulative installed capacity is better utilized and stabilized, ensuring a more consistent electricity supply and facilitating the integration of variable renewable energy sources such as solar PV.

This scenario aligns with Malawi's Integrated Energy Plan, which targets a 75% annual increase in Battery Energy Storage System (BESS) capacity starting from 2025. The plan builds upon an existing installed BESS capacity of 20 MW, with new additions accumulating each year to support the rapid expansion of solar PV which also grows at 75% annually and to stabilize the national grid.

BESS capacity begins accumulating in 2025 at 0.02 GW (20 MW) and increases steadily each year, reaching approximately 4.3 GW by 2050. This reflects the 75% annual growth target, ensuring that storage scales alongside variable renewable generation. The parallel growth of solar PV, which begins in 2035 and reaches roughly 0.6 GW by 2050, demonstrates a coordinated deployment strategy designed to manage intermittency and maximize renewable utilization.

These results illustrate a deliberate strategy to pair solar PV with BESS to mitigate intermittency, while using storage not only to support solar integration but also to stabilize hydropower output and enhance grid resilience. The scenario also supports a gradual reduction in reliance on imported power and fossil fuels, advancing Malawi's energy security and decarbonization goals.

However, it is important to note that a 75% annual increase in BESS capacity is somewhat over-ambitious, particularly when considering the substantial investment, operational, and maintenance costs associated with large-scale battery deployment. In practice, such aggressive growth may be constrained by funding limitations, supply chain bottlenecks, and the need for complementary grid infrastructure to fully utilize the added storage.

3. Conclusion and Policy Insights

The results indicate that Malawi's current electricity pathway (BAU) which is characterised by heavy reliance on hydropower (approximately 0.55 GW of installed capacity supplying over 80% of generation, equivalent to ~8–10 PJ annually) and coal-fuel backup is increasingly unsustainable in the face of climate variability and rapidly rising demand. While this pathway appears less capital-intensive at only 25.3% in the short term, it perpetuates supply insecurity, exposure to fuel price volatility, and delayed decarbonisation.

By contrast, renewable energy expansion, particularly solar PV, delivers substantial long-term benefits. Model results show that scaling solar PV to 1.5–2.0 GW by 2040, supplying 12–15 PJ of electricity annually, can reduce system-wide emissions by over 60% relative to BAU and lower average generation costs to below USD 0.08/kWh. However, these benefits are only realised when supported by adequate system flexibility. Without storage, high solar penetration leads to increased curtailment and reliability risks.

From a policy perspective, the Government of Malawi should prioritise a coordinated renewables-plus-storage strategy. This involves scaling up solar PV alongside a phased and financially realistic deployment of Battery Energy Storage Systems, for example 300–500 MW (1.2–2.0 GWh) of BESS by 2040, rather than pursuing overly aggressive annual capacity expansion targets that may strain public finances and grid infrastructure.

References

Malawi Vision 2063

Malawi National Energy Policy 2023

Malawi Integrated Energy Plan 2022

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